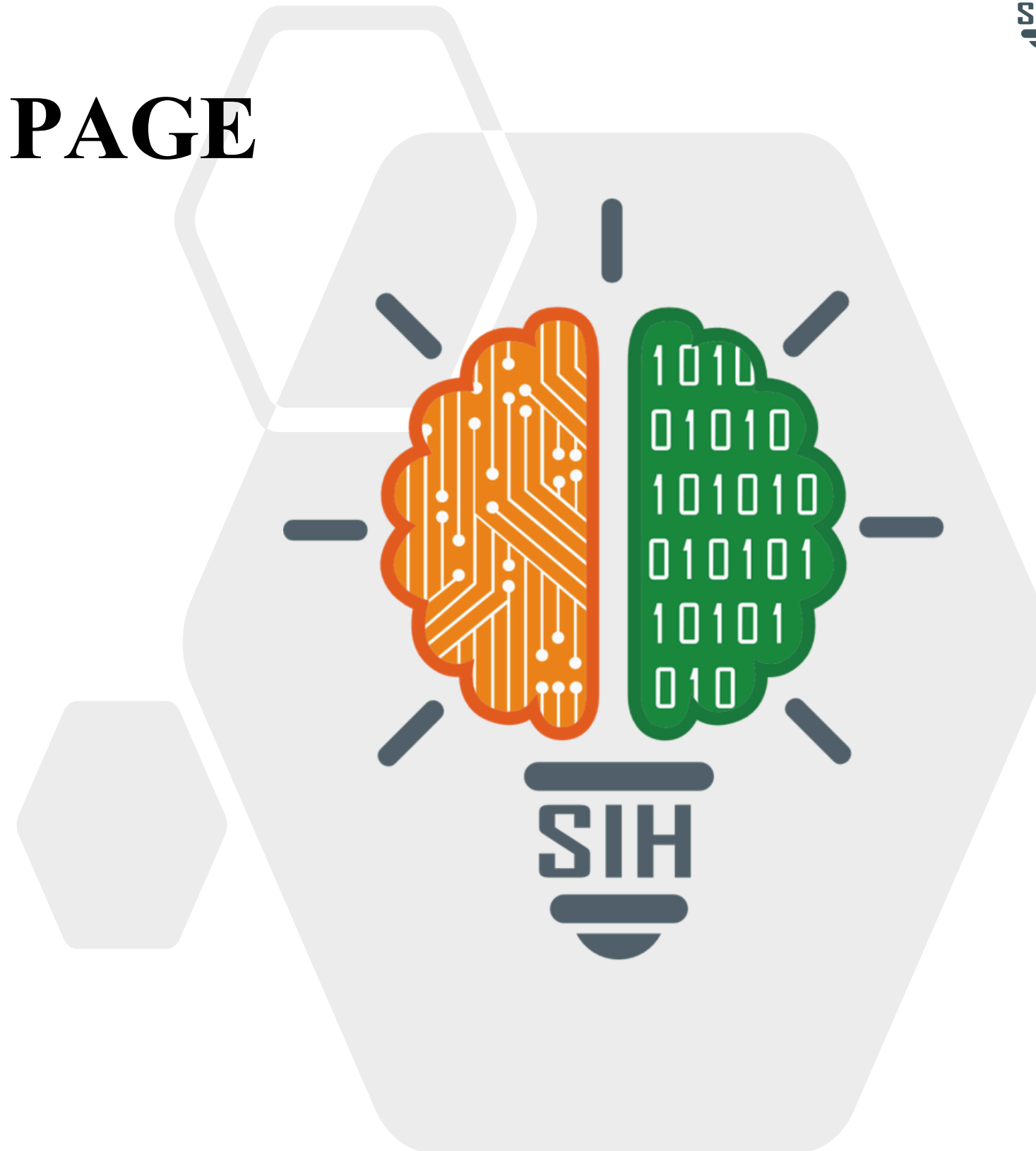


TITLE PAGE

- **Problem Statement ID – 26117**
- **Problem Title –**
Sovereign On-Premise Agentic AI Workbench
using Open-Weight Multimodal LLMs for
Confidential Industrial Work
- **Theme - Smart Automation**
- **PS Category- Software**
- **Team Name - BharatByte**



Enterprise AI capability without sending enterprise data to the cloud.

Our Solution



Private by Design

Runs completely on-premise using the organization's own GPU — zero external calls.



Agentic + Multi-Model

Automatically routes each task to the right open-weight model and uses local tools to plan, execute and verify.



Multimodal → Real Deliverables

Understands scanned documents, handwriting, P&IDs and images, then produces verified Word / Excel / PPT / Code grounded in local SOPs.

Before

Uploading to the cloud



- ❌ Data leaves your environment
- ❌ Higher privacy & security risks
- ❌ Depends on internet connectivity

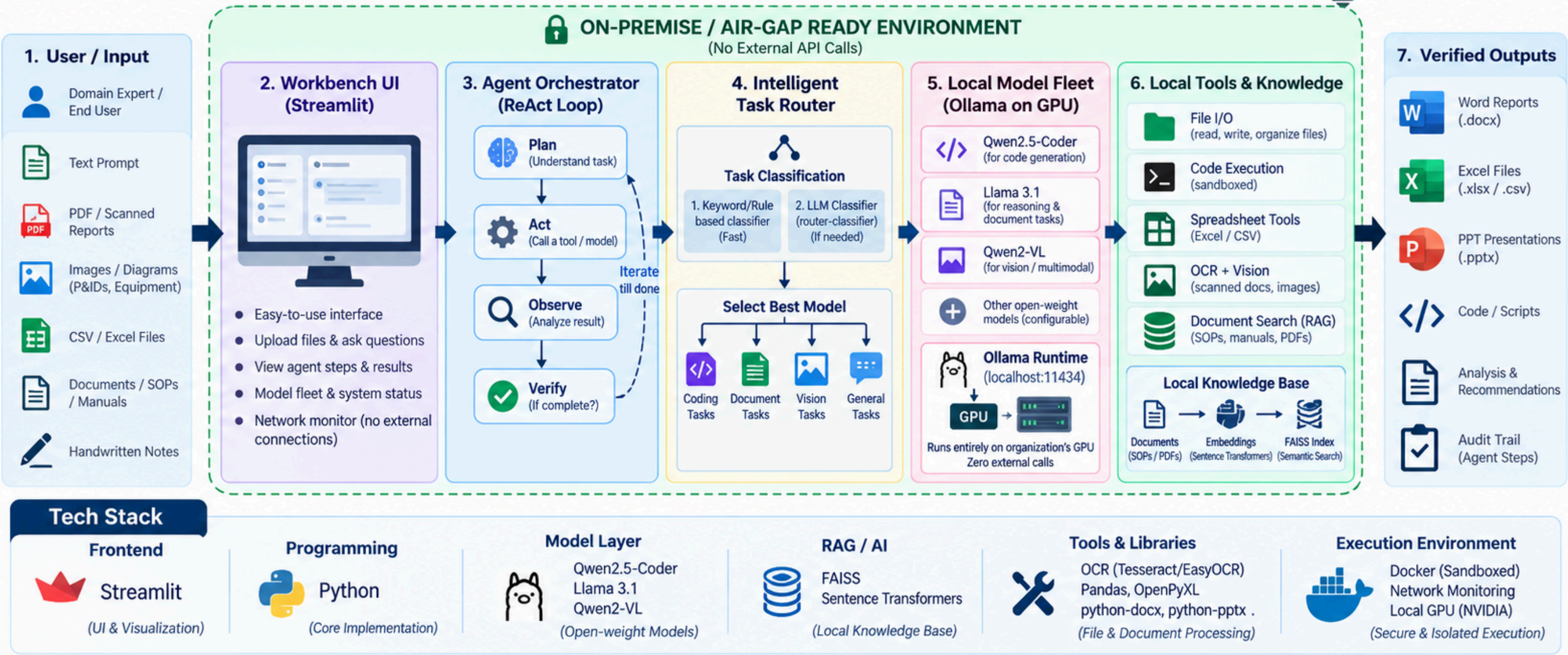
FROM
RISK
TO
CONTROL

After

On-Prem GPU Server



- ✅ Data stays within your infrastructure
- ✅ Stronger privacy & security
- ✅ Works independently of the internet



FEASIBILITY & VIABILITY

Deployable today • Risks identified • Clear path to production

A practical, scalable and secure solution built on open-source technology



IMPACT & BENEFITS

Enabling Secure, Productive and Self-Reliant Workspaces

Same AI capability. For the people who build and protect the nation.



Target Audience

Professionals who handle sensitive information and need a secure AI alternative



Refinery / PSU Engineers & Inspectors

Process data, inspection reports, maintenance records, SOPs



Defence-Linked Manufacturing Units

Design documents, test reports, compliance data



Government Offices

Policy drafts, internal reports, citizen data, administrative work



IT / Security Teams

Organizations with no sanctioned alternative to public AI tools

“ For teams who can't use public AI tools, but still need the productivity boost. ”



Key Benefits

Real value for real-world users



Security & Compliance

Removes the actual failure mode — confidential data leaving via public AI tools.

- ✓ Keeps data on-premise
- ✓ Meets regulatory requirements
- ✓ Builds trust for sensitive domains



Productivity

Cuts manual hours on approval notes, report summaries, internal tooling and analysis.

- ✓ Faster document understanding
- ✓ Automates routine tasks
- ✓ More time for critical engineering work



Adoption & Accessibility

Engineers get a copilot without breaking policy or falling behind cloud-tool users.

- ✓ Familiar, easy-to-use interface
- ✓ Works offline / on internal networks
- ✓ Drives AI adoption in restricted environments



Broader Impact



Stronger National Security

Critical and sensitive data stays within this country.



Operational Efficiency

Higher productivity across PSUs and government offices.



Cost Savings

No recurring API costs; leverages existing infrastructure.



Self-Reliance (Atmanirbhar Bharat)

Built on open-source models and local infrastructure.



Scalable to Other Sectors

Applicable to other regulated industries with similar constraints.

Research Papers

- Yao et al. (2022), ReAct: Synergizing Reasoning and Acting in Language Models — basis for the Plan → Act → Observe → Verify agent loop
- Lewis et al. (2020), Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks — grounds outputs in local SOPs/manuals via RAG
- Johnson, Douze & Jégou (2017), Billion-Scale Similarity Search with GPUs — FAISS, used for the local vector index

Model & Tooling Documentation

- Qwen2.5-Coder Technical Report (Alibaba Qwen Team, 2024) — coding-task model
- The Llama 3 Herd of Models (Meta AI, 2024) — reasoning/document-task model
- Qwen2-VL Technical Report (Alibaba Qwen Team, 2024) — vision/multimodal model
- Ollama documentation — local GPU model-serving runtime